



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2019 HSC Course Assessment Task 1

12th of December, 2018

General instructions

- Working time – 55 minutes
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.

QUESTIONS 1-12

- Commence each **new section** on a **new booklet**
- Commence each new question on a **new page**. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER:

BOOKLETS USED:

Class (please ✓)

- ☐ Mr Ireland
☐ Dr Jomaa
☐ Ms Lee

Marker's use only.

SECTION	A	B			C		D		
QUESTION	1 – 5	6	7	8	9	10	11	12	Total
MARKS	$\overline{5}$	$\overline{4}$	$\overline{5}$	$\overline{3}$	$\overline{7}$	$\overline{5}$	$\overline{5}$	$\overline{7}$	$\overline{41}$

Section A

Mark your answers on the multiple choice section in the answer booklet.

Marks

1. A quadratic expression with zeros of $4 + i$ and $4 - i$ is: **1**
- (A) $x^2 - 8x + 17$
- (B) $x^2 + 8x + 17$
- (C) $x^2 - 8x - 17$
- (D) $x^2 + 8x - 17$
2. Let $z = 4 + i$. What is the value of $\overline{iz}$? **1**
- (A) $-1 - 4i$
- (B) $-1 + 4i$
- (C) $1 - 4i$
- (D) $1 + 4i$
3. Let $z = 3 - i$. What is an expression for $\frac{1}{z}$? **1**
- (A) $\frac{3}{4} + \frac{1}{4}i$
- (B) $\frac{3}{8} + \frac{1}{8}i$
- (C) $\frac{3}{10} + \frac{1}{10}i$
- (D) $\frac{3}{2} + \frac{1}{2}i$
4. Given $3x^3 + 6x - 5 = 0$ has roots α, β and γ , what is the value of $\alpha^3 + \beta^3 + \gamma^3$? **1**
- (A) 5
- (B) 9
- (C) 15
- (D) -1

5. If ω is a complex cube root of unity, which of the following results is **NOT** correct? **1**
- (A) ω^2 is the other complex root
- (B) $\omega^2 + \omega = -1$
- (C) $\omega^6 - \omega = \omega^2$
- (D) $\bar{\omega}$ is also a root

End of Section A.

Examination continues overleaf.

Section B

Write your answers in the writing booklet supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (4 Marks)	Commence a NEW page.	Marks
-----------------------------	----------------------	--------------

- | | | |
|-----|--|----------|
| (a) | Express $\sqrt{24 - 70i}$ in the form $a + ib$. | 2 |
| (b) | Hence solve $z^2 - (1 - i)z - 6 + 17i = 0$. | 2 |

Question 7 (5 Marks)	Commence a NEW page.	Marks
-----------------------------	----------------------	--------------

- | | | |
|-----|---|----------|
| (a) | Given that $\sin 3\theta = 3 \sin \theta \cos^2 \theta - \sin^3 \theta$ and $\cos 3\theta = \cos^3 \theta - 3 \sin^2 \theta \cos \theta$,
prove that $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$ | 1 |
| (b) | Hence, or otherwise, find θ where $x^3 - 3x^2 - 3x + 1 = 0$ and $x = \tan \theta$ if $0 \leq \theta \leq \pi$ | 2 |
| (c) | Hence prove that $\tan^2 \frac{\pi}{12} + \tan^2 \frac{5\pi}{12} = 14$. | 2 |

Question 8 (3 Marks)	Commence a NEW page.	Marks
-----------------------------	----------------------	--------------

- | | | |
|-----|--|----------|
| (a) | If α, β, γ are zeros of the polynomial $x^3 - px^2 + qx - r$,
prove $\beta^2\gamma^2 + \gamma^2\alpha^2 + \alpha^2\beta^2 = q^2 - 2pr$ | 3 |
|-----|--|----------|

Section C

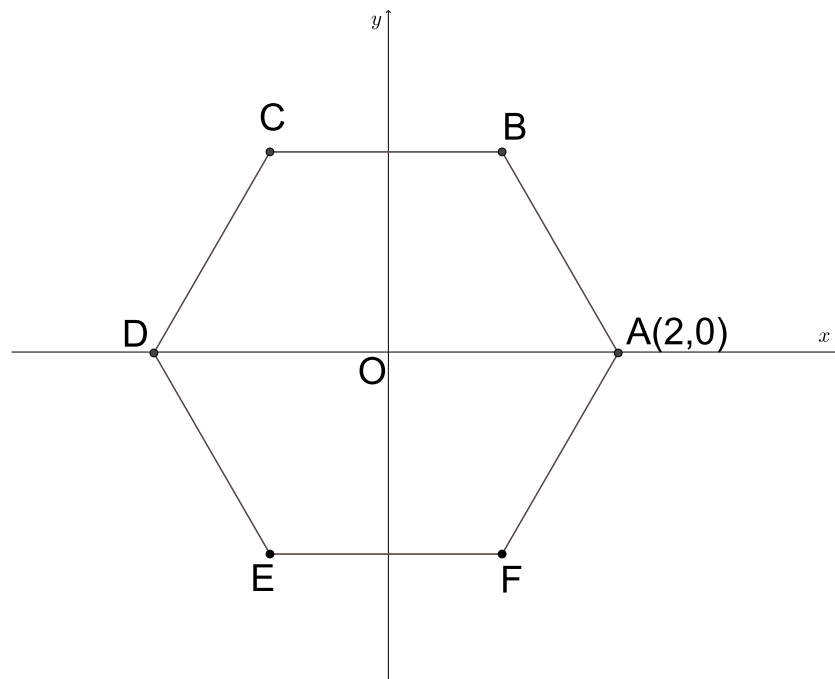
Write your answers in the writing booklet supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (7 Marks)

Commence a NEW page.

Marks

$ABCDEF$ is a regular hexagon drawn on an Argand diagram with vertex A at the point $(2, 0)$. O is the centre of the hexagon.



- Sketch the region within the hexagon that satisfies $|z| \geq 1$ and $-\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{3}$. 2
- Find, in the form $|z - c| = R$, the equation of the circle through the points O , B and F . 1
- Find the complex numbers represented by the points C and E . 2
- The hexagon is rotated anticlockwise about the origin through an angle of $\frac{\pi}{2}$. Express in the form $r(\cos \theta + i \sin \theta)$ the complex numbers represented by the new positions of C and E . 2

Question 10 (5 Marks)

Commence a NEW page.

Marks

If α is a zero of the polynomial $P(z) = az^2 + bz + c$, which has real coefficients a, b and c ,

- (a) Show that $\bar{\alpha}$ is a zero of $P(z)$ **2**
- (b) Find the monic polynomial of least degree, with integer coefficients, having $1 - i$ as a root of multiplicity 2. **3**

Section D

Write your answers in the writing booklet supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (5 Marks)

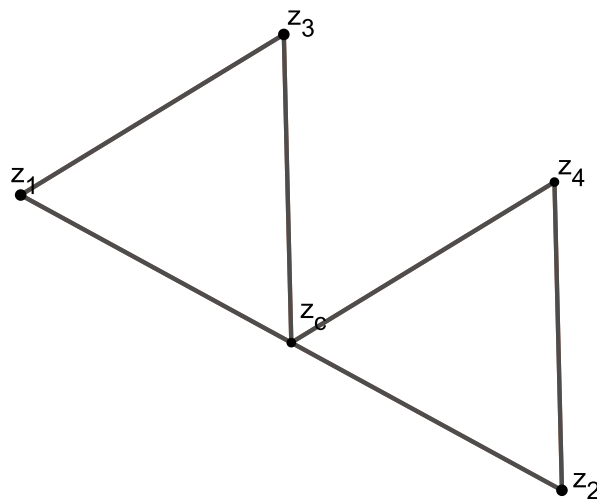
Commence a NEW page.

Marks

- (a) Solve the equation $z^4 - 1 = 0$ where z is a complex number **2**
- (b) Hence find all solutions of the equation $z^4 = (z - 1)^4$. **3**

Question 12 (7 Marks)

Commence a NEW page.

Marks

Two complex numbers z_1 and z_2 are given on the Argand diagram above.

z_c is the complex number representing the midpoint of z_1z_2 .

z_3 is a complex number such that

$$(z_3 - z_1)^2 - (z_3 - z_1)(z_c - z_1) + (z_c - z_1)^2 = 0$$

- (a) Find $\frac{z_3 - z_1}{z_c - z_1}$. Express in the form $r(\cos \theta + i \sin \theta)$ **2**
- (b) Hence show that z_1, z_c, z_3 are the vertices of an equilateral triangle. **2**
- (c) z_4 is the point such that z_c, z_2, z_4 are the vertices of an equilateral triangle **2**
 Show that $(z_4 - z_3)^2 + (z_4 - z_3)(z_1 - z_3) + (z_1 - z_3)^2 = 0$

(d) What is the nature of triangle $z_1z_4z_3$?

1

End of paper.

BLANK PAGE

BLANK PAGE

Answer sheet for Section A

Mark answers to Section A by fully blackening the correct circle, e.g. “●”

STUDENT NUMBER:

Class (please ✓)

☐ Mr Ireland

☐ Dr Jomaa

☐ Ms Lee

1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

5 – ☐ A ☐ B ☐ C ☐ D

Section 11

1. (A)
2. (A)
3. (C)
4. (A)
5. (C)

Section B

Question 6

2 a) Let $\sqrt{24-70i} = a+ib$

$$a^2 - b^2 = 24$$

$$a^2 + b^2 = \sqrt{24^2 + 70^2} = 74 \quad \text{①} \quad \text{a, b opposite signs}$$

$$\text{①} + \text{②} \quad 2a^2 = 98$$

$$a^2 = 49$$

$$a = \pm 7$$

$$b^2 = 49 - 24 = 25$$

$$b = \pm 5$$

$$\therefore \sqrt{24-70i} = \pm(7-5i)$$

2 b) $z^2 - (1-i)z - 6+17i = 0$

$$z = \frac{(1-i) \pm \sqrt{(1-i)^2 - 4(-6+17i)}}{2}$$

$$= \frac{(1-i) \pm \sqrt{-2i + 24 - 68i}}{2}$$

$$= \frac{(1-i) \pm \sqrt{24-70i}}{2}$$

$$= \frac{(1-i) \pm (7-5i)}{2}$$

$$= 4-3i, -3+2i$$

Question 7

1 a) $\tan 3\theta = \frac{\sin 3\theta}{\cos 3\theta}$

$$= \frac{3\sin\theta\cos^2\theta - \sin^3\theta}{\cos^3\theta - 3\sin^2\theta\cos\theta} \times \frac{\cos^3\theta}{\cos^3\theta} \quad (\cos\theta \neq 0)$$

$$= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

2 b) Let $x = \tan\theta$

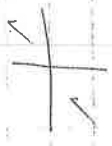
$$\tan 3\theta = \frac{3x - x^3}{1 - 3x^2}$$

If $\tan 3\theta = 1$,

Then $1 = \frac{3x - x^3}{1 - 3x^2}$

$$1 - 3x^2 = 3x - x^3$$

$$\text{i.e. } x^3 - 3x^2 - 3x + 1 = 0$$



Now $\tan 3\theta = 1$

for $3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$ if $0 \leq \theta \leq \pi$

$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{3\pi}{4}$ $0 \leq \theta \leq 3\pi$

∴) Sum of roots are at a time:

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} + \tan \frac{3\pi}{4} = 3$$

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4 \quad \text{Since } \tan \frac{3\pi}{4} = -1$$

Product of roots:

$$\tan \frac{\pi}{12} \tan \frac{5\pi}{12} \tan \frac{3\pi}{4} = -1$$

$$\tan \frac{\pi}{12} \tan \frac{5\pi}{12} = +1$$

$$\tan^2 \frac{\pi}{12} + \tan^2 \frac{5\pi}{12} = \left(\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} \right)^2 - 2 \tan \frac{\pi}{12} \tan \frac{5\pi}{12}$$

$$= 4^2 - 2(+1)$$

$$= 16 - 2$$

$$= 14$$

⑧

$$x^3 - px^2 + qx - r = 0$$

Replace x with $\sqrt{x}$ for eqn. i root $\alpha^2, \beta^2, \gamma^2$.

$$x\sqrt{x} - px + q\sqrt{x} - r = 0$$

$$x\sqrt{x} + q\sqrt{x} = px + r$$

$$x(x+q)^2 = (px+r)^2$$

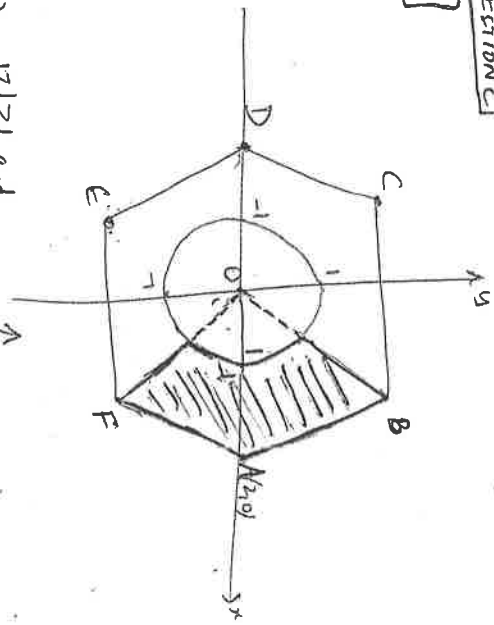
$$x(x^2 + 2qx + q^2) = p^2x^2 + 2prx + r^2$$

$$x^3 + 2qx^2 + q^2x = p^2x^2 + 2prx + r^2$$

$$x^3 + (2q - p^2)x^2 + (q^2 - 2pr)x - r^2 = 0$$

Sum of roots two at a time:

$$q^2 - 2pr$$



(a) $|z| \geq 1$ and $-\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{3}$.

#

✓

(b) Since $AB = AO = AF = 2$

$\therefore A$ is the centre & radius = 2

$|z - 2| = 2$

#

✓

(c) $\vec{OC} = \cos \frac{2\pi}{3} \times \vec{OA}$

$= \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)(2 + 0i)$

$= -1 + i\sqrt{3}$

C is $-1 + i\sqrt{3}$.
(or $2 \cos \frac{2\pi}{3}$)

✓

Similarly,

$\vec{OE} = \cos \left(-\frac{2\pi}{3}\right) \times \vec{OA}$

$= \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)(2 + 0i)$

$= -1 - i\sqrt{3}$

E is $-1 - i\sqrt{3}$.

#

(or $2 \cos \left(-\frac{2\pi}{3}\right)$)

(d) $\vec{OC}' = i \vec{OC}$

$= i(-1 + i\sqrt{3})$

$= -\sqrt{3} - i$

$C' = -\sqrt{3} - i$
 $= 2 \cos \left(-\frac{5\pi}{6}\right)$ (or $2 \cos \frac{7\pi}{6}$)

Also, $\vec{OE}' = i \vec{OE}$

$= i\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)$

$= \frac{\sqrt{3}}{2} - \frac{i}{2}$

$E' = \frac{\sqrt{3}}{2} - \frac{i}{2}$
 $= 2 \cos \left(-\frac{\pi}{6}\right)$

#

SECTION C

(Q10)

(a) $P(z) = az^2 + bz + c$, (a, b, c all real)

Let α be a zero of $P(z)$, i.e. $P(\alpha) = 0$.

Now, $P(\bar{\alpha}) = a(\bar{\alpha})^2 + b\bar{\alpha} + c$

$= a\bar{\alpha}^2 + b\bar{\alpha} + c$, as $(\bar{z})^n = \overline{z^n}$ ✓

$= \overline{a\alpha^2 + b\alpha + c}$, as $\overline{c\bar{z}} = \overline{c\bar{z}}$ for $\left[\begin{matrix} \frac{1}{a} \\ \frac{1}{b} \\ \frac{1}{c} \end{matrix} \right]_{a, b, c \in \mathbb{R}}$

$= \overline{a\alpha^2 + b\alpha + c}$, as $\overline{\overline{w + z}} = \overline{w + z}$

$= \overline{P(\alpha)}$

$= \overline{0}$
 $= 0$ $\therefore \bar{\alpha}$ is also a zero of $P(z)$. # ✓

(b). Since $1-i$ is a root, so is $1+i$ (because our polynomial has integer i.e. real coefficients).

But $1-i$ is a root of multiplicity 2, and $P(z)$ is monic,

Hence:

$P(z) = [z - (1-i)]^2 [z - (1+i)]^2$ ✓

$= [z - (1-i)]^2 [z - (1-i)]^2$

$= [z - (1-i)] [z - (1-i)] \times [z - (1+i)] [z - (1+i)]$

$= [z - (1-i)^2 + 1] [z - (1+i)^2 + 1] = (z^2 - 2z + 2)^2$

$= (z - 1)^4 + 2(z - 1)^2 + 1$

$= z^4 - 4z^3 + 6z^2 - 4z + 1 + 2z^2 - 4z + 2 + 1$

$P(z) = z^4 - 4z^3 + 8z^2 - 8z + 4$. ✓ #

Section D

Question 11

(a) $z^4 - 1 = 0$

$z^4 = 1$

$\cos 4\theta = \cos 2k\pi$ where $k \in \mathbb{Z}$

$4\theta = 2k\pi$

$\theta = \frac{k\pi}{2}$

For principal arguments, $-\pi < \frac{k\pi}{2} \leq \pi$

$-2 < k \leq 2$

$k = -1: z_1 = \cos \frac{-\pi}{2} = -i$

$k = 0: z_2 = \cos 0 = 1$

$k = 1: z_3 = \cos \frac{\pi}{2} = i$

$k = 2: z_4 = \cos \pi = -1$

(b) $z^4 = (z - 1)^4$

$\left(\frac{z}{z-1}\right)^4 = 1$

Let $y = \frac{z}{z-1}$

$zy - y = z$

$z(y-1) = y$

$z = \frac{y}{y-1}$

$= \frac{-i}{-i-1} = \frac{1}{1-i} = \frac{1}{1-i} \times \frac{1+i}{1+i} = \frac{1+i}{2}$

$= \frac{1+i}{2} \times \frac{1-i}{1-i} = \frac{1-i}{2}$

$= \frac{1+i}{2}, \frac{1-i}{2}, \frac{1}{2}, \frac{1}{2}$

Question 12

(a) $(z_3 - z_1)^2 - (z_3 - z_1)(z_c - z_1) + (z_c - z_1)^2 = 0$

Multiply by $(z_3 - z_1 + z_c)$, we obtain:

$$(z_3 - z_1 + z_c - z_1)((z_3 - z_1)^2 - (z_3 - z_1)(z_c - z_1) + (z_c - z_1)^2) = 0$$

$$\therefore (z_3 - z_1)^3 + (z_c - z_1)^3 = 0$$

$$\therefore (z_3 - z_1)^3 = -(z_c - z_1)^3$$

$$\left(\frac{z_3 - z_1}{z_c - z_1}\right)^3 = -1 = \cos \pi$$

$$\therefore \frac{z_3 - z_1}{z_c - z_1} = \cos \frac{\pi}{3}$$

$$\left| \frac{z_3 - z_1}{z_c - z_1} \right| = \left| \cos \frac{\pi}{3} \right| = 1 \quad \therefore |z_3 - z_1| = |z_c - z_1|$$

$\therefore \triangle ABC$ is an isosceles triangle

$$\angle ABE = \angle BAH = \frac{\pi - \frac{\pi}{3}}{2} = \frac{\pi}{3} \quad (\text{Equal angles opposite equal sides and angle sum of a triangle})$$

$\therefore \triangle ABC$ is an equilateral triangle

$$\angle z_1 z_3 z_4 = \frac{2\pi}{3}$$

$$\frac{z_4 - z_3}{z_1 - z_3} = \cos \frac{2\pi}{3}$$

$$\therefore \left(\frac{z_4 - z_3}{z_1 - z_3}\right)^3 = \cos 2\pi = 1$$

$$\therefore (z_4 - z_3)^3 = (z_1 - z_3)^3$$

$$\therefore (z_4 - z_3)^3 - (z_1 - z_3)^3 = 0$$

$$(z_4 - z_3)(z_4^2 + z_4 z_3 + z_3^2) - (z_1 - z_3)(z_1^2 + z_1 z_3 + z_3^2) = 0$$

Since $z_4 - z_1 \neq 0$

$$(z_4 - z_3)^2 + (z_4 - z_3)(z_1 - z_3) + (z_1 - z_3)^2 = 0$$

